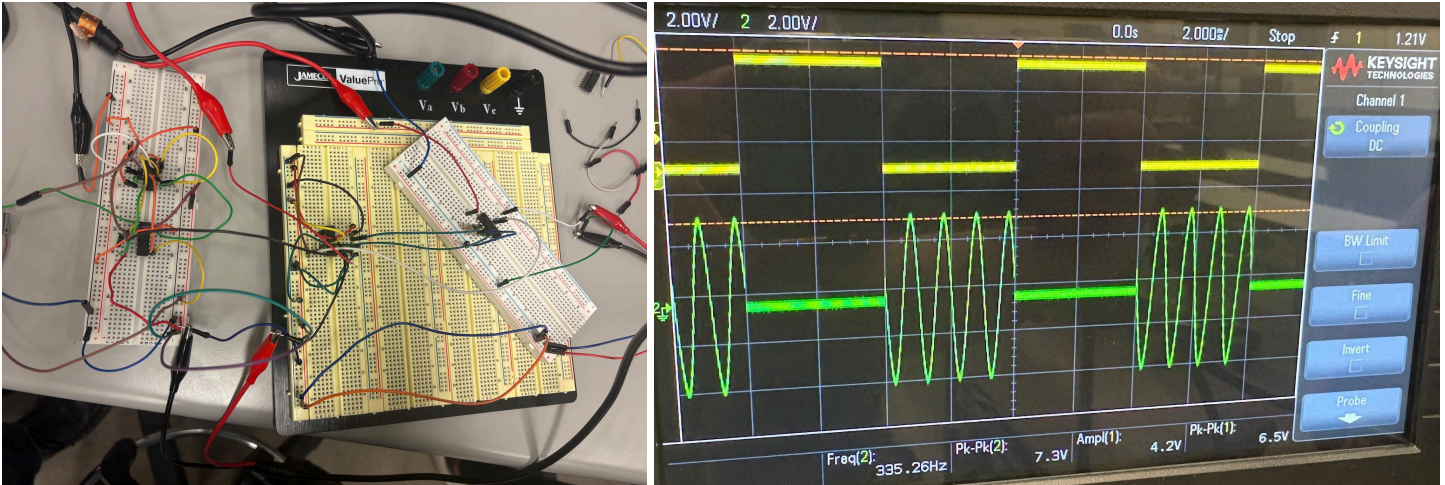


ETE 4351L Data Communication and Networking

Lab 8

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March 17, 2026



OBJECTIVES

- Create a pseudo-random data (PRNG) generator using a Linear Feedback Shift Register (LFSR)
- Understand the operation of an analog multiplexer (MUX)
- Create and examine a circuit which implements Amplitude Shift Keying (ASK) modulation and demodulation
- Reinforce use of laboratory equipment

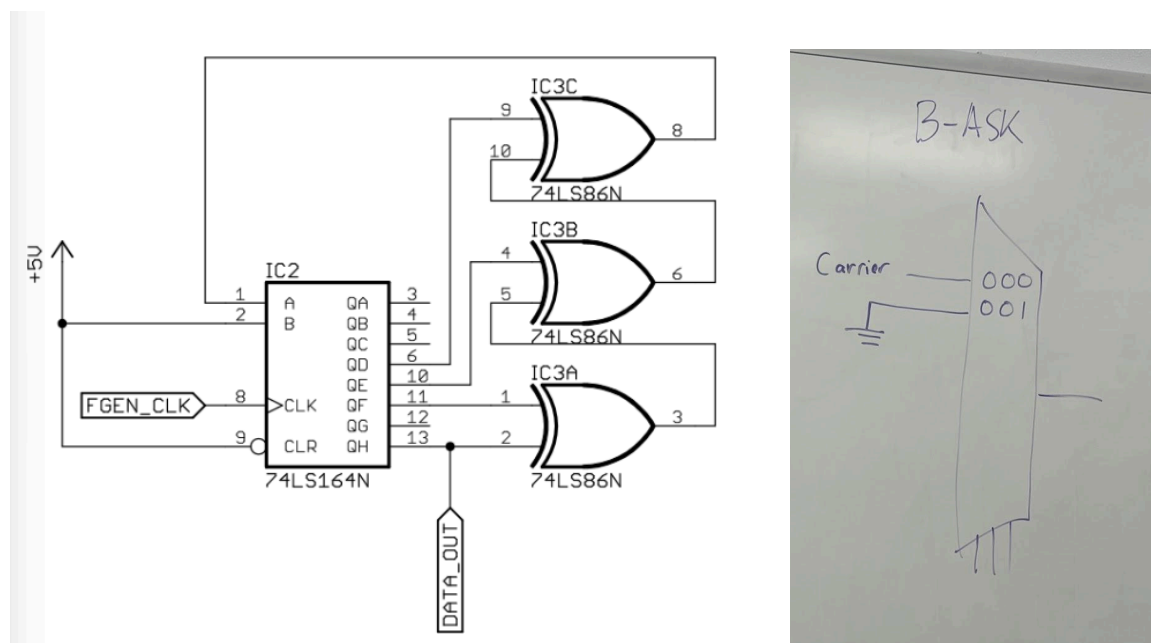
Introduction

In this lab Amplitude Shift Keying (ASK) is used as a method of transmitting digital information using an analog carrier signal. ASK is a form of amplitude modulation, where the information is digital and the amplitude of the carrier changes in discrete levels as opposed to continuously. In binary ASK (or 2 ASK) a 1 bit is represented by the presence of the carrier while a 0 bit is represented by the absence of the carrier. This is also known as on-off keying.

This lab also examines how digital signals relate to frequency content and bandwidth. Since digital data resembles a square wave it contains a fundamental frequency and infinite harmonics. In practice and reality only a limited number of these frequency components can be transmitted as we cannot have an infinite bandwidth, meaning that we must approximate the signal using its fundamental frequency. The relationship between data rate, clock frequency and bandwidth is also considered showing that the frequency content of the data is related to half the clock frequency. Additionally this lab demonstrates how modulation shifts the frequency content of the signal, producing sum and difference frequencies around the carrier. This leads to an understanding of the minimum bandwidth required to transmit ASK signals and the importance of limiting bandwidth to avoid interference with other signals.

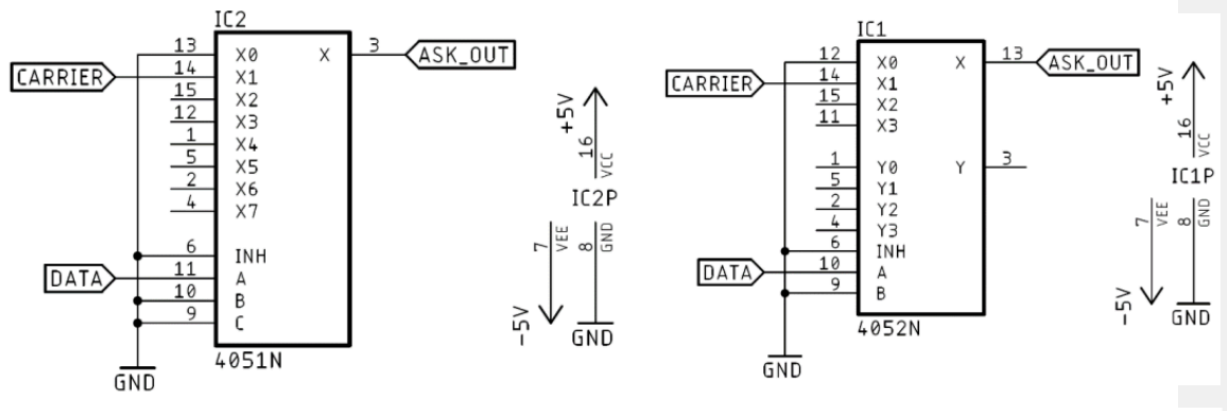
Procedure:

1. Build the circuit of Figure 1. This is the PRNG circuit.



IMG 1. Linear Feedback Shift Register

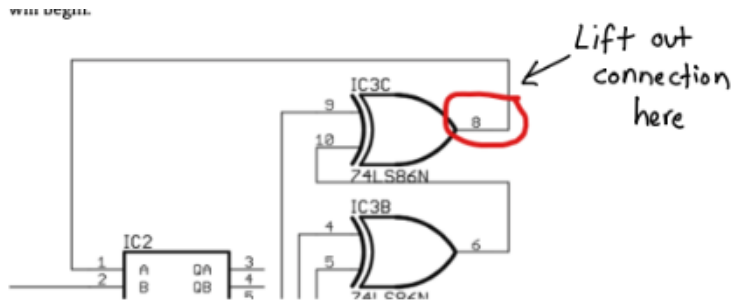
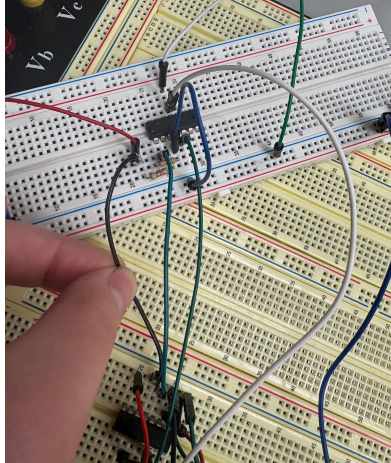
IMG 2. Diagram from class



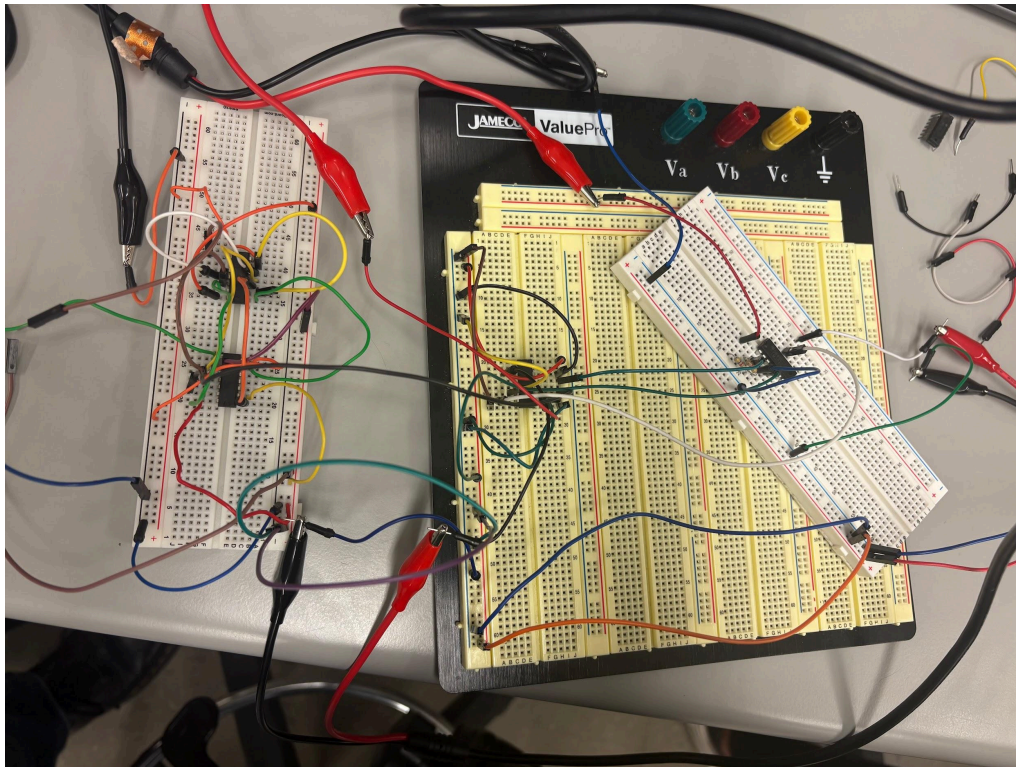
IMG 2. ASK modulator circuits, CD4051 (left) and CD4052 (right)

Notice there are three power pins on the 4051/4052: Vcc, Vee, and GND. There is a positive, negative, and ground. The remainder of circuit is left over from lab 7, and we are simply just editing the output of 4051.

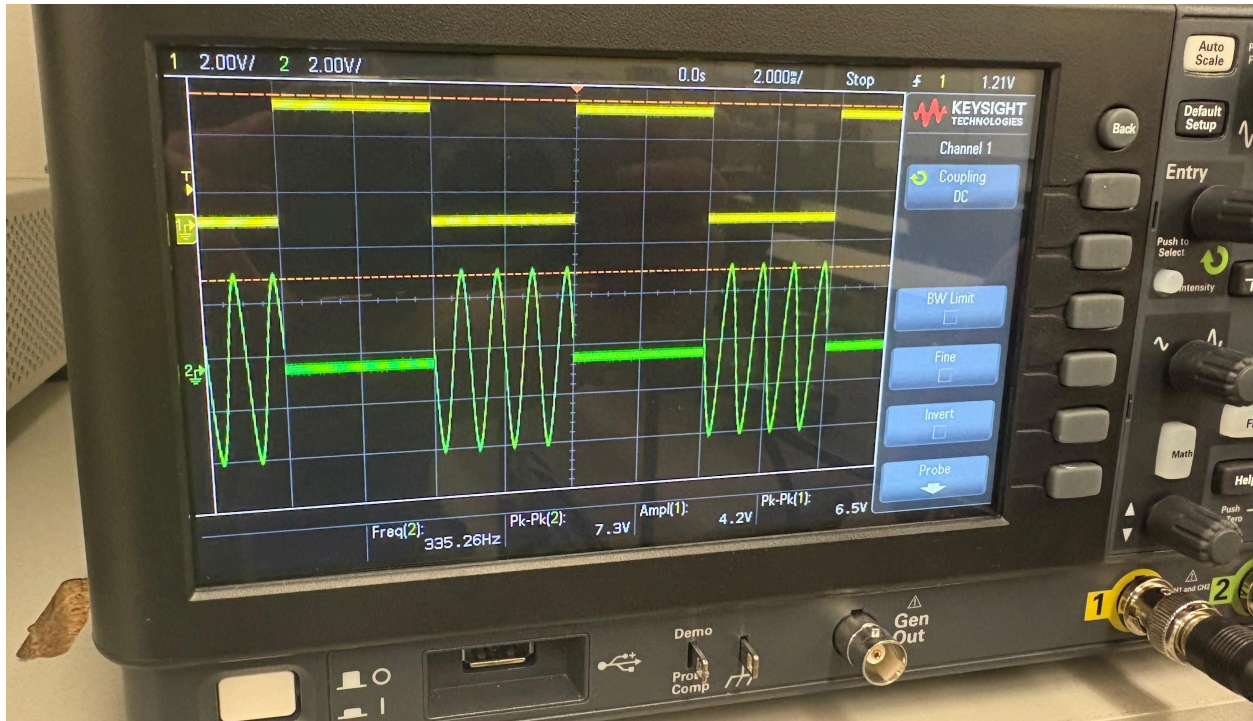
2. You must initially connect pin A of the shift register to +5V to clock in some '1' bits. Then, move the connection back to the feedback from the XOR gates (as shown in the schematic).
3. Set the clock generator to 1 KHz square wave, 5Vpp, +2.5VDC offset, High-Z.
4. Capture a screenshot showing the clock input and random data output on the same plot. Note: you need to "seed" the random number generator circuit by temporarily removing the feedback between the XOR output and shift register input. You can remove the connection from the XOR side, and "tap" it to +5V for a second and then put it back to the XOR. This will allow at least a few '1' bits to be clocked into the shift register and the random data sequence will begin.



IMG 3 and 4. Pertaining to this segment of the circuit, moving the black wire grounds 180 phase shift which makes it change by amplitude.



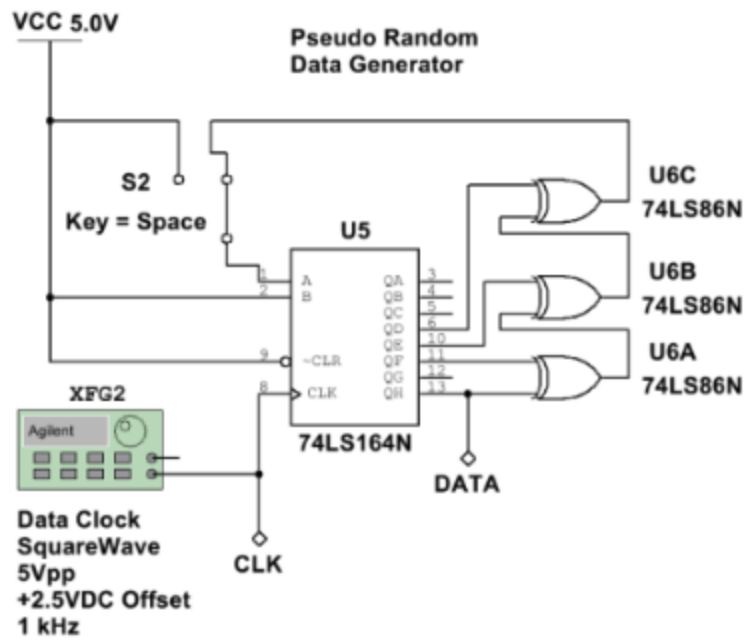
IMG 5. The circuit



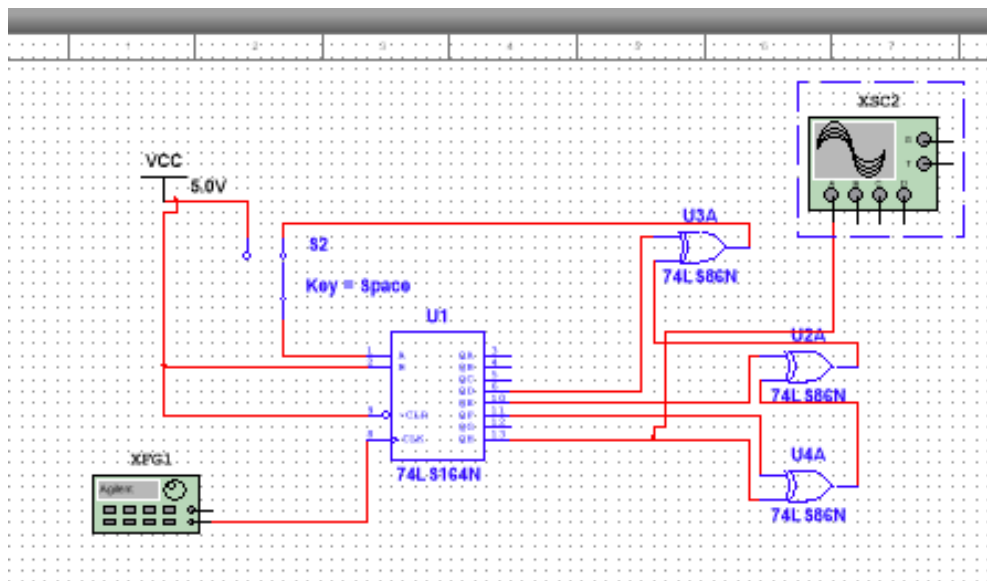
IMG 6. *Scope output*

Multism

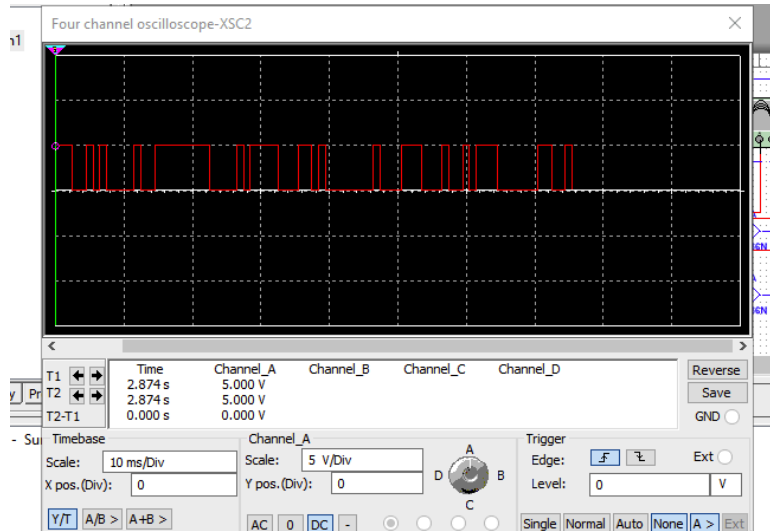
1. In Multisim, build the PRNG circuit. After starting the simulation, you must move the switch over to the left as shown in the figure to clock a few '1' bits into the shift register, and then the switch can be moved back to the position containing the XOR feedback output.
2. Verify that the circuit is operational before moving on. You should see a random pattern of ones and zeros in the "DATA" output.



IMG 7. Multisim circuit schematic

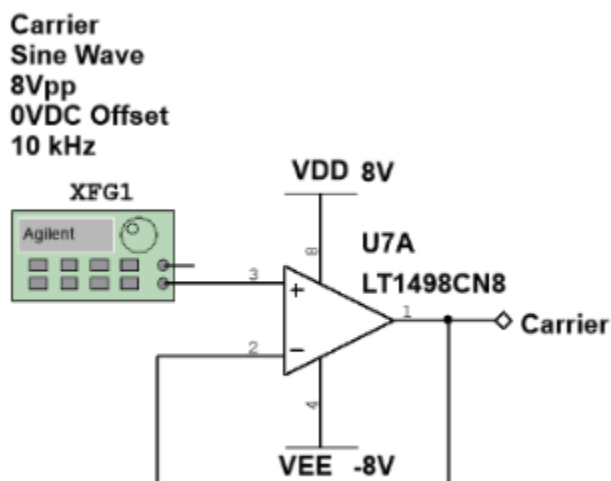


IMG 8. Multisim circuit

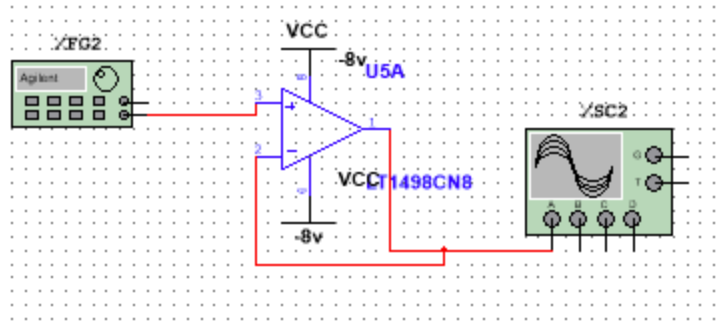


IMG 9. *Multisim circuit output*

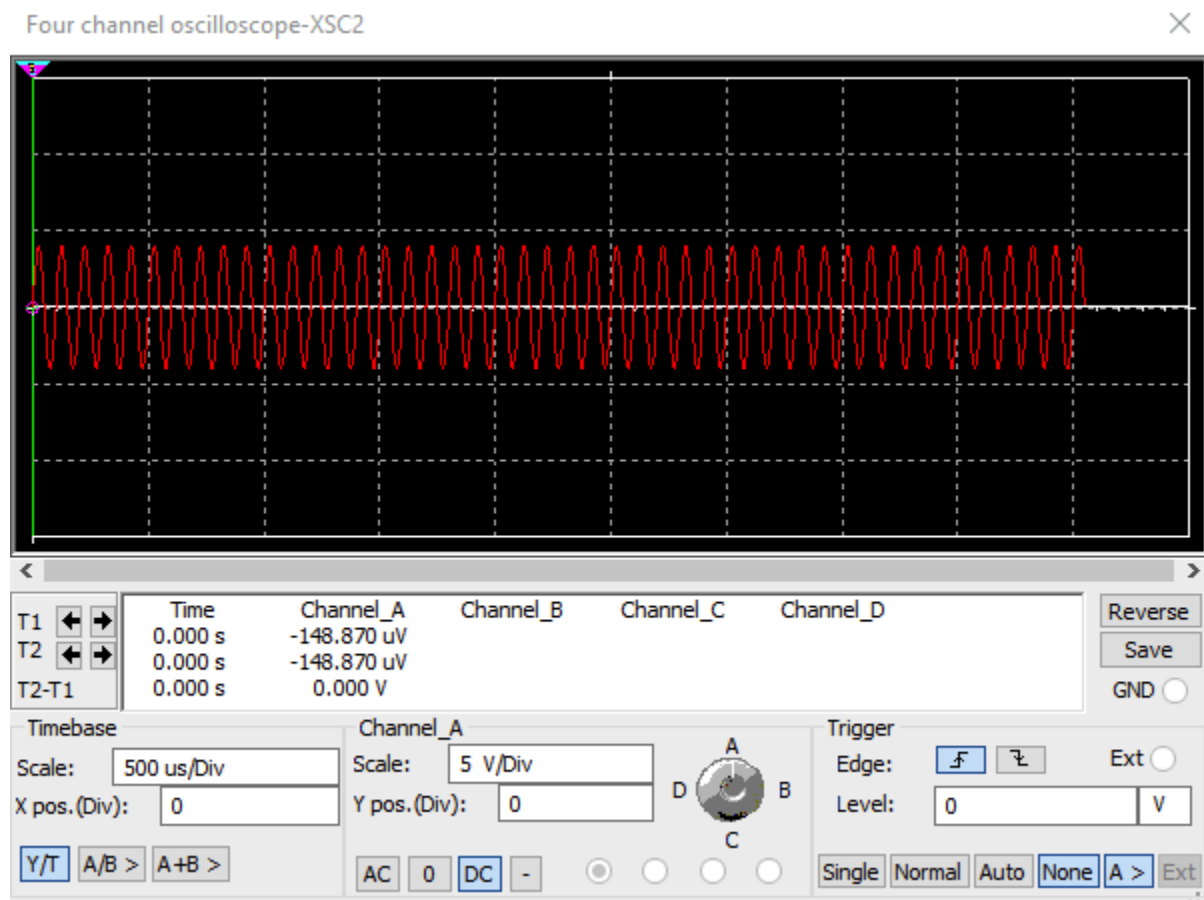
3. Now, create the carrier signal using another function generator and a buffer amplifier. Verify the carrier output port/ connector is displaying the correct signal.



IMG 10. *Carrier signal*

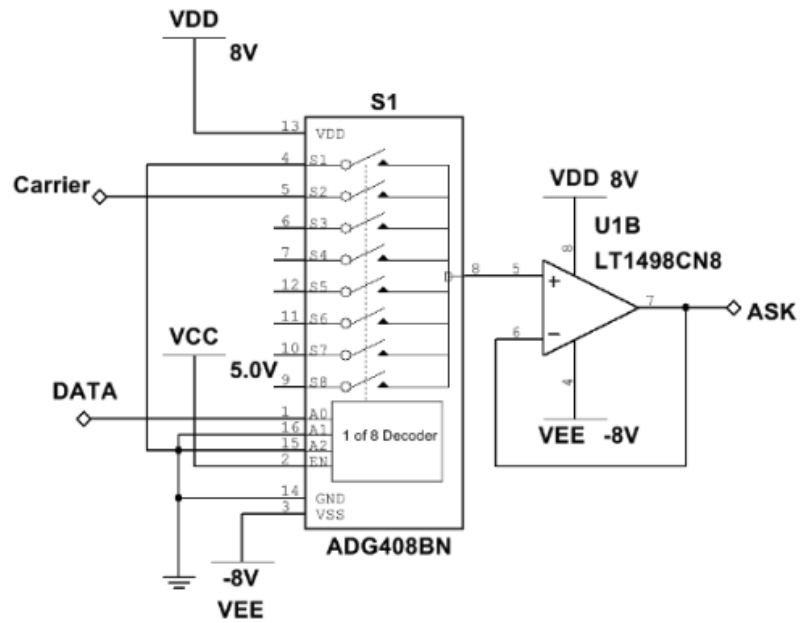


IMG 11. Multisim circuit of carrier

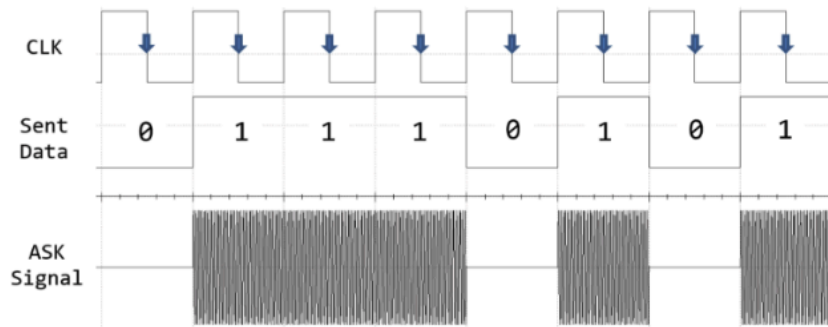


IMG 12. Multisim circuit output

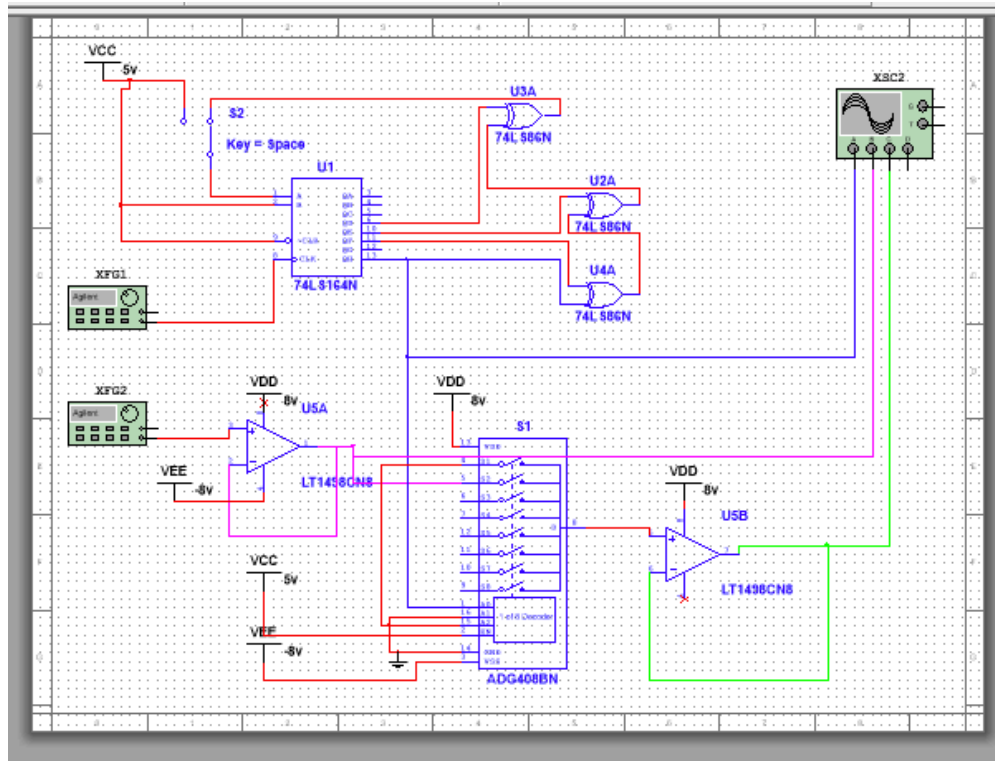
4. Create the ASK modulator using an analog MUX and another buffer. Verify the ASK output by comparing the clock, data and ASK signal on the same plot as shown below. Note the power supply voltages here of +8v, -8v and ground.



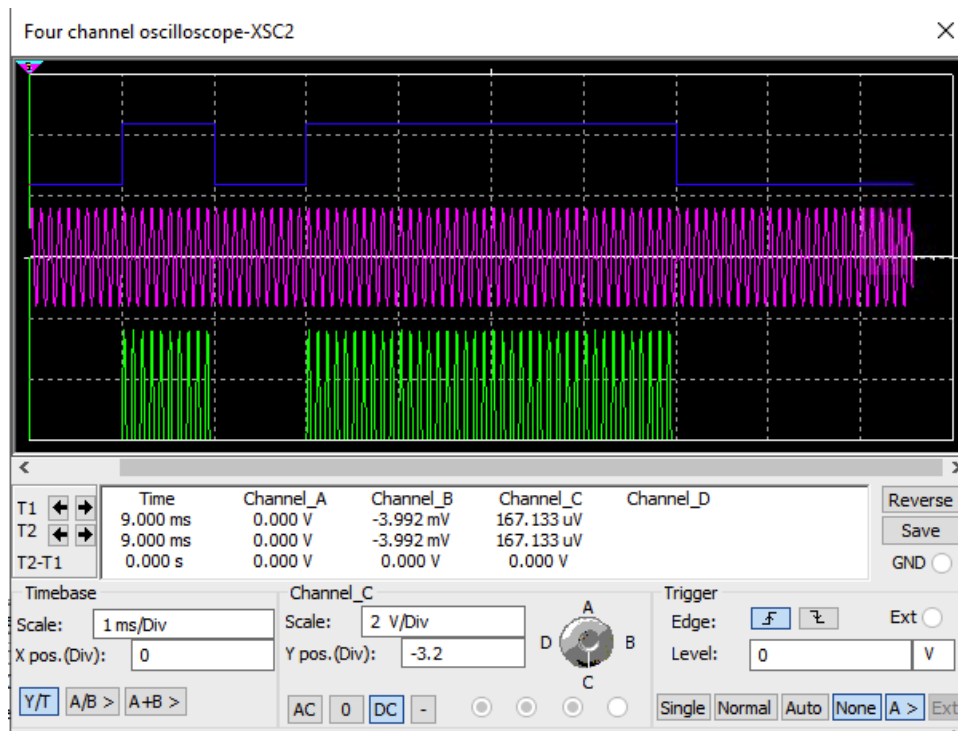
IMG 13. *ASK modulator using analog MUX*



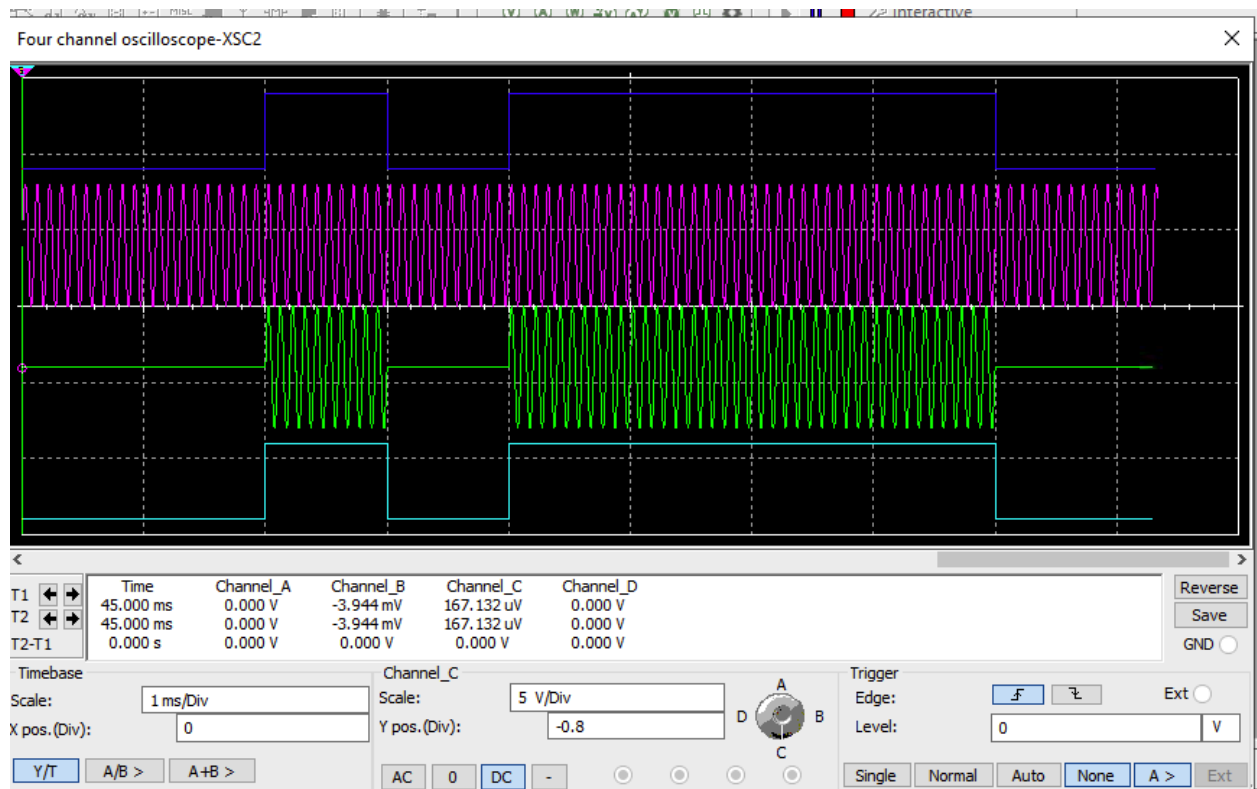
IMG 14. *ASK modulated signal vs data expected output*



IMG 15. Circuit

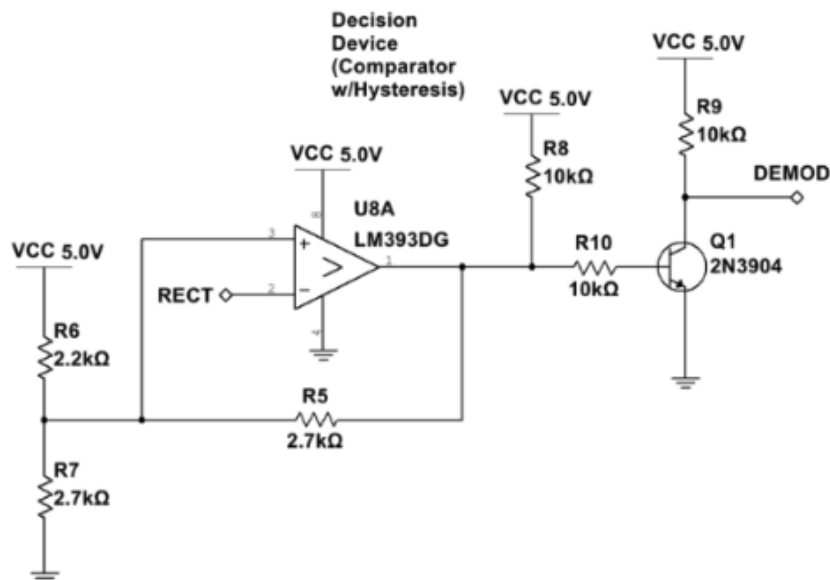


IMG 16. Circuit output (blue is data, pink is carrier, green is ASK output)

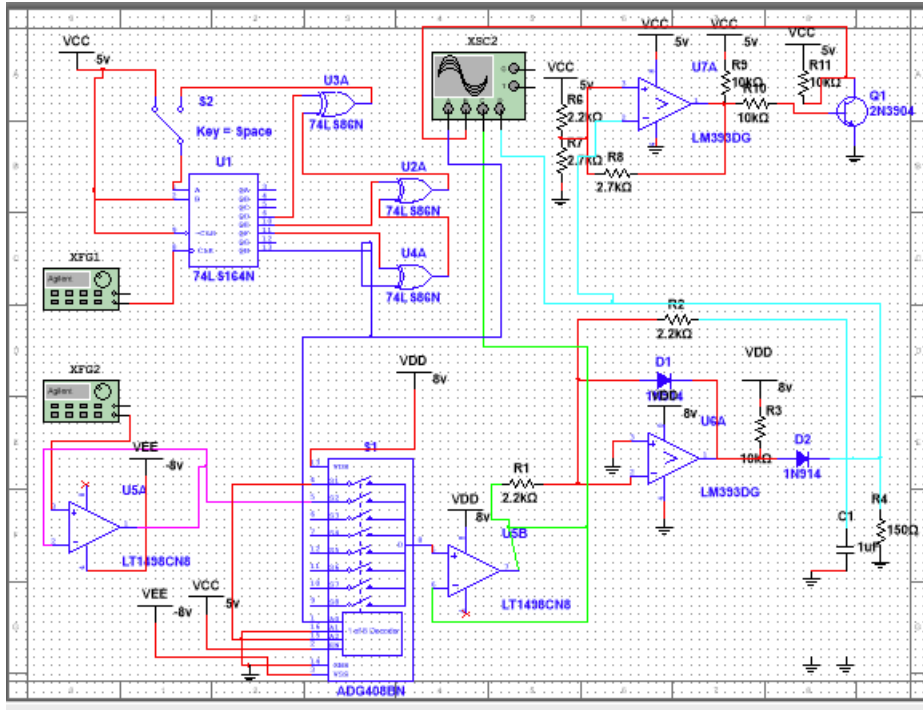


IMG 19. Circuit with envelope detector output

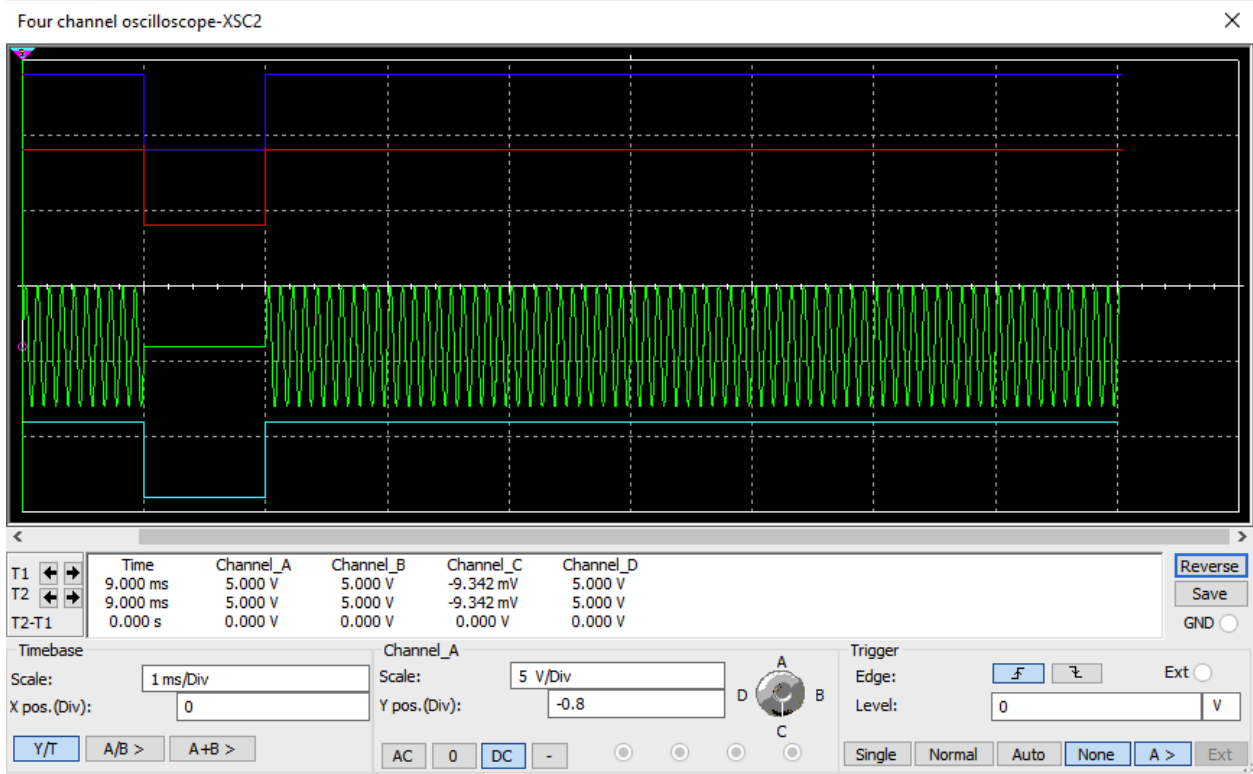
6. Build the decision device, which will make decisions about whether a 1 bit or 0 bit is being sent.



IMG 20. Decision device

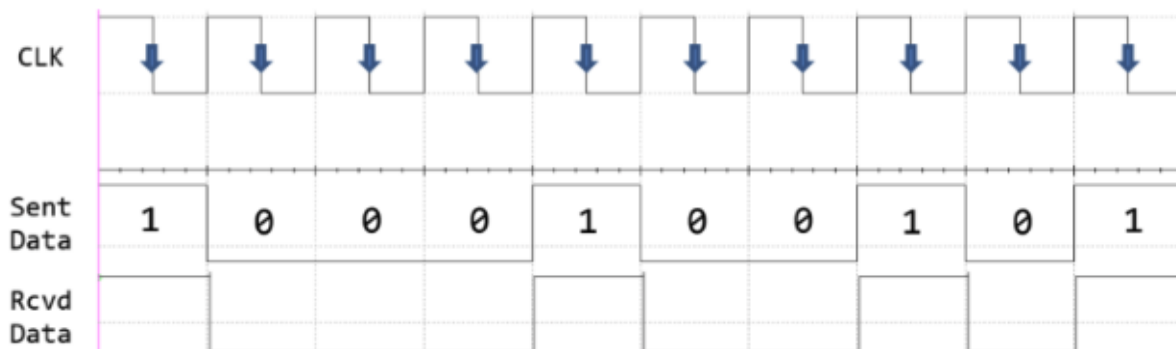


IMG 21. Circuit with decision device.



IMG 22. Decision device. Red is after demodulation, dark blue is the original data, light blue is the demodulated data.

Compare the clock original data bitstream from PRNG, and the final demodulated output at receiver (DEMOD) as shown below:



IMG 23. *Expected demodulated output*

Our output (image 19) matches the expected demodulated output (image 20). The data is successfully demodulated and the ASK signal clearly changes based on the data from the random number generator.

Theoretical analysis

- For ASK, explain how the sine wave carrier is modulated by the data stream. What does the ASK signal look like when a '1' bit is being sent? How about a '0' bit?

In amplitude shift keying (ASK) the amplitude of a sine wave carrier is changed based on the digital data signal. When a 1 bit is transmitted the carrier is present and the sine wave appears at its full amplitude. When a 0 bit is transmitted the carrier is absent meaning that the output is close to 0v. This is also called on off keying since the carrier is either turned on or off depending on the data.

- What data sequence results in the worst-case bandwidth?

The worst case data sequence for bandwidth occurs when the data alternates continuously, such as 101010 or 01010101. This is because there is a lot of sharp transitions, meaning that a higher frequency content is needed and thus requiring the largest bandwidth.

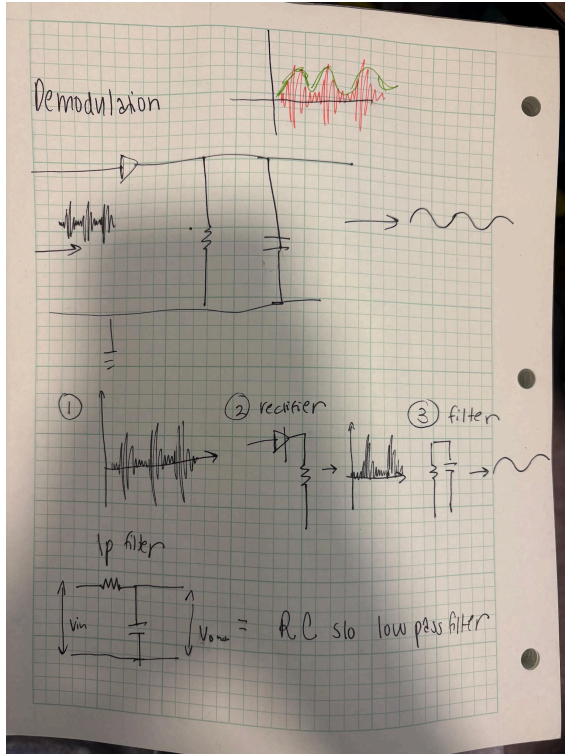
- Find the minimum passband bandwidth BW of the ASK modulated circuit in this implementation (which occurs when only the fundamental frequency is included).

The minimum passband bandwidth of the ASK signal is equal to twice the fundamental frequency of the data signal. Since the fundamental frequency of the data is half of the clock frequency, the bandwidth becomes equal to the data rate. Therefore the minimum bandwidth required is equal to the data bit rate.

- Can we accurately decode the data at the receiver without any knowledge of the clock timing (phase)? At what part of each bit cycle should we sample the data relative to the sender's clock (not the carrier)? Maybe discuss a flip flop to show how the flip flop output can transition only $\frac{1}{2}$ as fast as the clock that feeds the flip flop.

No, we cannot accurately decode the data at the receiver without knowledge of the clock timing. The receiver must know when to sample the signal in order to determine correctly whether a 1 or 0 is being transmitted. The signal should be sampled near the middle of each period, where the signal is most stable and least affected by transitions. The importance of timing can actually be understood by considering a flip flop. This is because a flip flop updates its output only on clock edges, meaning its output can change at most once per clock cycle. As a result the data output can only transition at half the rate of the clock in the worst case. This shows that the data transitions occur at predictable timing relative to the clock and that sampling has to consider this to avoid errors. If the receiver does not know the clock timing or samples at the wrong time (like during a transition) it may incorrectly interpret the signal and cause errors.

- Consult the internet to explain how a basic envelope detector circuit works (just a diode, resistor, and capacitor).



We learned about envelope detectors in ETE 3351. This circuit consists of a diode, resistor and capacitor. The diode and resistor function as a rectifier. The rectifier is used to extract the envelope (shape) of the signal by allowing current to flow only in one direction. After the diode the capacitor stores the voltage from the rectified signal and then smooths it over time. There is also a low pass filter (from the resistor and capacitor), removing the high frequency carrier.

- Explain the operation of a basic comparator. How is a comparator with hysteresis different from the basic comparator?

A basic comparator compares two input voltages and outputs either a high or low signal depending on which input is greater. If the input voltage is above a certain threshold, the output is high. If it is below the threshold, the output is low. A comparator with hysteresis is different from a basic comparator because it improves this behavior. It does so by using two different threshold voltages (one for switching high and one for switching low). This prevents noise from causing rapid switching at the threshold and results in a cleaner, more stable output.

Conclusion

In this lab Amplitude Shift Keying (ASK) was used to transmit digital data by varying the amplitude of a carrier signal. A pseudo random data signal (from the circuit which was built in the previous lab) was generated and used to control the presence or absence of the carrier, demonstrating how digital information can be represented using analog signals. The behavior of the ASK signal was observed showing us that a '1' corresponds to the carrier being present while a '0' corresponds to its absence. The lab also demonstrated the relationship between digital data and frequency content, by showing that digital signals require bandwidth based on their harmonic components. By considering only the fundamental frequency, the minimum bandwidth required for transmission was determined. Additionally, the effects of modulation were observed including the creation of sum and difference frequency components around the carrier.

Overall, this lab demonstrated how ASK modulation works, how bandwidth requirements are determined, and how digital signals can be effectively transmitted and analyzed in the frequency domain.